Elastoplasticity

Ruiyu Ou

August 11, 2026

Contents

0.1 The Elastoplasticity Problem

0.1.1 Description and Connection with Cold Spray

Plasticity is the study of materials that undergo irreversible, permanent deformations under large loads.

Cold spray deposition results from micron sized metal particles impacting a target substrate at supersonic speeds. Plastic and thermomechanical effects are the dominant physics in cold spray deposition. Under the viscoplastic deformation and thermomechanical behaviors, the metal particles undergo shear thinnign then metallurgically binds the target substrate. Understanding the plastic processes is necessary for modeling cold spray deposition.

0.1.2 The Mathematical Problem

Plasticity is inherently an experimental science. The mathematical theories of plasticity are formulated to represent experimental findings and physical models, and such is more phenomenological. The physical models of plasticity is concerned with how plasticity occurs.

The physical material of interest will completely dictate inelastic processes. The materials of interest can vary greatly from fairly regular metals, to soil types, and even rocks and concrete. Respectively, some modes of fault: slip bands, compactification of soil or voids and shearing, and opening and closing of cracks. [5]

To describe the onset and accumulations of permanent deformation, known as yielding, we need a criterion for the transition from elastic to plastic deformation. From physical experiments, the transition from elastic to plastic may not be obvious, if such a point even exists in reality. There are then 2 main types of constitive models for plastic deformation. One assuming the existence of a yield point, and one that does not. Here we present the former. [4]

Suppose there exists a function of all internal state variables called the yield function – $F(q)$. We denote the collection of all state variables with q. We assume the following.

$$\text{if } F(q) < 0, \text{ elastic deformation for all such q. Plastic otherwise.} \tag{1}$$

Naturally, the manifold given by $F(q) = 0$ represent the point of transition from elastic to plastic deformation. The manifold is known as the yield surface.

The specific $F(q)$ will depend on the model.

As all thermodynamical models go, we assume the existence of a free energy ψ, depending on the elastic strain and internal state variables. We then obtain the following equations for stress and hardening.

$$\sigma = \bar{\rho}\frac{\partial \psi}{\partial \varepsilon^e} \ ; \ A = \bar{\rho}\frac{\partial \psi}{\partial \alpha} \tag{2}$$

Where σ is the stress, ε^e is the elastic strain, A the thermomechanical hardening process, and α the internal state variables.

A complete thermomechanical description of a plastic material also needs the evolution laws for the internal state variables. We present the following equations for the time evolution of the plastic strain ε^p, and internal variables α.

$$\dot{\varepsilon}^p = \dot{\gamma}N(\sigma, A) \ ; \ \dot{\alpha} = \dot{\gamma}H(\sigma, A) \tag{3}$$

$$N = \frac{\partial f}{\partial \sigma} \ ; \ H = -\frac{\partial f}{\partial A} \tag{4}$$

Where $\dot{\gamma}$ is a proportionality parameter known the plastic multiplier, N known as the flow vector, H known as the hardening modulus, adn they are derived from a plastic flow potential f.

The evolution laws are complemented with a set of sensible physics known as the loading/unloading conditions, which may be of a familiar form from some contact problems.

$$F \leq 0 \ ; \ \dot{\gamma} \geq 0 \ ; \ \dot{\gamma}F = 0 \tag{5}$$

Recall the criterion for the onset of plastic deformation Eq 1. If the conditions are purely elastic, no thermomechanical processes occur, and the evolution of internal state variables are null. In this case we can trivially force $\dot{\gamma} = 0$. If the conditions are elastoplastic, there are some form of thermomechanical process so $\dot{\gamma} \geq 0$ and $F \geq 0$. However, physically the time evolution of the elastoplastic body is continuous in phase space. It must continuously move from 1 internal state corresponding to elastic deformation $F < 0$ to plastic deformation $F \geq 0$. With a trivial connectedness consideration, it must begin plastically deforming at $F = 0$, and the internal state variables will evolve with it as the current stress is equivalent to the yield stress. Which then naturally gives us that $\dot{\gamma} \geq 0$, and $F = 0$ under plastic deformation.

All together, we have these equations that describe plasticity.

$$\text{Additivity of strain: } \varepsilon = \varepsilon^e + \varepsilon^p \tag{6}$$

$$\text{Free energy: } \psi(\epsilon^e, \alpha) \tag{7}$$

$$\text{Constitutive relations of stresses: } \sigma = \bar{\rho}\frac{\partial \psi}{\partial \varepsilon^e} \; ; \; A = \bar{\rho}\frac{\partial \psi}{\partial \alpha} \tag{8}$$

$$\text{Yield function: } F(\sigma, A) \tag{9}$$

$$\text{Evolution of internal variables: } \dot{\varepsilon}^p = \dot{\gamma}N \; ; \; \dot{\alpha} = \dot{\gamma}H \tag{10}$$

$$\text{Loading/unloading conditions: } F \leq 0 \; ; \; \dot{\gamma} \geq 0 \; ; \; \dot{\gamma}F = 0 \tag{11}$$

0.2 FreeFEM Implementation

0.2.1 Physical Model

We concern ourselves with the model of a 50um by 75um block of copper impacting a rigid wall at 500m/s.

We proceed with von Mises' J_2 based associative flow with Johnson-Cook model. [3]

The yield function of the plasticity model is given by the difference of the von Mises stress and the Johnson-Cook yield stress.

$$F(q) = \sigma_{vm} - \sigma_Y \tag{12}$$

$$\sigma_{vm} = \sqrt{\frac{3}{2}s : s} \; ; \; s = \sigma - \frac{1}{3}\operatorname{Tr}(\sigma)I \tag{13}$$

$$\sigma_Y = (A + B\bar{\varepsilon}_p^{n_{JC}})(1 + C\ln(\dot{\epsilon}_p/\dot{\epsilon}_0))(1 - T^{*m}) \; ; \; T^* = \frac{T - T_0}{T_m - T_0} \tag{14}$$

$$\bar{\varepsilon}_p = \int |\dot{\varepsilon}_p| dt = \int \dot{\bar{\varepsilon}}_p dt \; ; \; \dot{\bar{\varepsilon}}_p = \sqrt{\frac{2}{3}\dot{\varepsilon}_p : \dot{\varepsilon}_p} \tag{15}$$

Where σ_{vm} is the von Mises stress, σ_Y the yield stress, $\bar{\varepsilon}_p$ is the equivalent/accumulated plastic strain.

For associative flow, the flow potential is equivalent to the yield function.

For the accumulation of plastic deformation and internal state variables, it will be given by the associative flow rule.

$$\dot{\varepsilon}_p = \dot{\gamma}\frac{\partial F}{\partial \sigma} = \dot{\gamma}\frac{3s}{2\sigma_{vm}} \tag{16}$$

$$\dot{\varepsilon} = \dot{\gamma}\frac{\partial F}{\partial A} = \dot{\gamma} \tag{17}$$

Fortunately for us, with von Mises' J_2 based Johnson-Cook associated flow plasticity, $\dot{\gamma}$ is equivalent to $\dot{\bar{\epsilon}}_p$.[1] This gives us a lot less to worry about in the return mapping algorithm later.

$$\dot{\bar{\varepsilon}}_p = \sqrt{\frac{2}{3}\dot{\varepsilon}_p : \dot{\varepsilon}_p} = \sqrt{\dot{\gamma}^2 \frac{3s : s}{2\sigma_{vm}^2}} = \dot{\gamma} \tag{18}$$

Where we use the loading/unloading criterion to drop absolute value.

The power delivered via the plastic process is given by the following equation.

$$\dot{W}_p = \sigma : \dot{\epsilon}_p \tag{19}$$

We suppose only a fraction of the energy is turned into heat, so the temperature change is given by the following.

$$\dot{T} = \chi \frac{s : \dot{\epsilon}_p}{\rho C_p} = \chi \dot{\bar{\epsilon}}_p \frac{\sigma_{vm}}{\rho C_p} = T_k \dot{\bar{\epsilon}}_p \sigma_{vm} \tag{20}$$

χ is Taylor-Quincy heat fraction. Where we define $T_k = \frac{\chi}{\rho C_p}$ and call it the temp fraction.

We lay out all the constants used and move towards the numerical implementation.

```
// Copper particle
real rho         = 8960.;         // Density
real E           = 117e9;         // Young's modulus
real nu          = 0.34;          // Poisson modulus
real A           = 90e6;          // 1st Johnson-Cook
    hardening parameter
real B           = 292e6;         // 2nd Johnson-Cook
    hardening parameter
real nJC         = 0.31;          // Johnson-Cook hardening
    strain exponent
real C           = 0.025;         // Johnson-Cook rate
    parameter
real m           = 1.09;          // Johnson-Cook thermal
    exponent
real dEps0       = 1.0;           // Reference strain rate

real Tm          = 1356.;         // Melting temp
real Cp          = 385.;          // Specific heat
real chi         = 0.9;           // Heat fraction
real Tk          = chi/(rho*Cp);  // Temp fraction
real T0     = 300.;               // Starting temp

real vimp   = -500.;      // Impact velocity
```

0.2.2 Elastic Portion only

We set up the model as a linear elastic problem, integrated over time to 50ns using Newark beta.

To avoid the cost of a contact algorithm, we idealize the circumstances and initiate the model right when the block hits the wall, and enforce a Dirichlet condition such that it doesn't penetrate it.

0.2.3 Plasticity with Linear Elasticity

We will adopt the backward euler return mapping as detailed in [2].

We present the incremental form of the system of equations.

$$\varepsilon^e_{n+1} = \varepsilon^{e\ trial}_{n+1} - \Delta\gamma[(1-\theta)N_n + \theta N_{n+1}] \tag{21}$$

$$\alpha_{n+1} = \alpha_n + \Delta\gamma[(1-\theta)H_n + \theta H_{n+1}] \tag{22}$$

$$F(\sigma_{n+1}, A_{n+1}) = 0 \tag{23}$$

$$\varepsilon^p_{n+1} = \varepsilon^{p\ trial}_{n+1} + \Delta\gamma[(1-\theta)N_n + \theta N_{n+1}] \tag{24}$$

$$\sigma_{n+1} = \sigma^{trial}_{n+1} - \Delta\gamma D^e : [(1-\theta)N_n + \theta N_{n+1}] \tag{25}$$

$$\sigma = D^e : \varepsilon^e \tag{26}$$

Where D^e is the elasticity tensor. θ is a parameter to tune the return mapping for desired performances. With $\theta = 1$ we obtain 1st order accuracy and unconditional convergence with the von Mises J_2 type yield surface.

The elastic predictor/return mapping algorithm steps are as follows.

- 1. Elastic trial step: given $\Delta\varepsilon$ and state variables at current time predict the state at the next time step.

$$\begin{aligned} \varepsilon^{e\ trial}_{n+1} &= \varepsilon^e_n + \Delta\varepsilon \\ \alpha^{trial}_{n+1} &= \alpha_n \\ \sigma_{n+1} = \bar{\rho}\frac{\partial F}{\partial \varepsilon^e}\ ;\ A_{n+1} &= \bar{\rho}\frac{\partial F}{\partial \alpha} \end{aligned}$$

- 2. Check plastic admissibility.

$$\begin{aligned} \text{If } F(\sigma^{trial}_{n+1}, A^{trial}_{n+1}) \le 0 \\ \text{Update by setting } (\cdot)_{n+1} = (\cdot)^{trial}_{n+1} \text{ and restart loop.} \end{aligned}$$

- 3. Return mapping algorithm. Solve the system.

$$\begin{bmatrix} \varepsilon^e_{n+1} - \varepsilon^{e\ trial}_{n+1} + \Delta\gamma[(1-\theta)N_n + \theta N_{n+1}] \\ \alpha_{n+1} - \alpha_n - \Delta\gamma[(1-\theta)H_n + \theta H_{n+1}] \\ F(\sigma_{n+1}, A_{n+1}) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- 4. Restart loop

Or simplified in plain english: 1. assume time step is fully elastic. 2. Check that it actually is elastic and accept if so. 3. If plastic, correct the behavior to match plastic constraints. 4. Loop to next time step.

We now explicitly write out all the incremental formulas in the return mapping algorithm.

$$
\begin{aligned}
\varepsilon_{n+1}^{p\ trial} &= \varepsilon_n^p \\
\varepsilon_{n+1}^{e\ trial} &= \varepsilon_n^e + \Delta\varepsilon = \varepsilon_{n+1} - \varepsilon_n^p \\
\bar{\varepsilon}_{n+1}^{p\ trial} &= \bar{\varepsilon}_n^{p\ trial} \\
\sigma_{Yn+1}^{trial} &= \sigma_{Yn} \\
s_{n+1}^{trial} &= 2\mu\varepsilon_{d\ n+1}^{e\ trial} \\
\varepsilon_{n+1}^{e} &= \varepsilon_{n+1}^{e\ trial} - \Delta\gamma\frac{3s_{n+1}}{2\sigma_{n+1}^{VM}} \\
\bar{\varepsilon}_{n+1}^{p} &= \bar{\varepsilon}_n^{p} + \Delta\gamma \\
s_{n+1} &= s_{n+1}^{trial} - 2\mu\Delta\gamma\frac{3s_{n+1}}{2\sigma_{n+1}^{VM}} \\
s_{n+1} &= \left(1 - \Delta\gamma\frac{3\mu}{\sigma_{n+1}^{VM\ trial}}\right)s_{n+1}^{trial} \\
\tilde{F}(\Delta\gamma) &= \sigma_{n+1}^{VM\ trial} - 3\mu\Delta\gamma - \tilde{\sigma}_Y(\Delta\gamma)
\end{aligned}
$$

0.2.4 Results

We first present the elastic results. At the end of those 50ns, we already see the block begin to rebound off the wall. We are definitely accumulating errors due to not using a contact algorithm.

References

[1] Dassault Systèmes, Providence, RI. *Johnson-Cook Plasticity*, 2024. Abaqus 2024 Documentation.

[2] Eduardo A. de Souza Neto, Djordje Perić, and David R. J. Owen. *Computational Methods for Plasticity: Theory and Applications*. John Wiley & Sons, Chichester, 2008.

[3] Gordon R. Johnson and William H. Cook. Fracture characteristics of three metals subjected to various strains, strain rates, temperatures and pressures. *Engineering Fracture Mechanics*, 21(1):31–48, 1985.

[4] Akhtar S. Khan and Sujian Huang. *Continuum Theory of Plasticity*. John Wiley & Sons, New York, 1995.

[5] Jacob Lubliner. *Plasticity Theory*. Macmillan, New York, 1990.